

STA5077Z

Unsupervised Learning

LECTURING TEAM

Miguel Rodó · Andrew Paskaramoorthy

DELIVERY

In person · R-based

COURSE

Overview

This course introduces statistical methods for identifying and analysing structure in multivariate datasets without a response variable. Topics include principal component analysis, multidimensional scaling, self-organising maps, ISOMAP, autoencoders and cluster analysis. The course carries 12 credits.

LECTURES

Schedule

Lectures are held in person according to the schedule below.

TIME

4–6 pm

DAYS

Mon · Wed · Fri, excluding 10 August

DATES

27 July–24 August

LECTURES 1–5

PD Hahn LT4

LECTURES 6–12

Hoerikwaggo LT1

Lectures	Topic
1–2	PCA and MDS
3–4	SOM and ISOMAP
5–6	Autoencoders
7–12	Cluster analysis

SUPPORT

Consultation sessions

Consultation arrangements will be communicated to the class during the course.

Content and practice material

LECTURE MATERIAL

Course resources

There is no prescribed textbook for this course. Slides, R scripts and other course material will be provided on Amathuba. All analyses and examples will be conducted using R.

CODING TUTORIALS

Practising implementation

Students will receive coding tutorials to practise implementing and applying the methods covered in the course using R.

THEORY PRACTICE

Preparing for tests and the exam

Additional theory-style questions will be provided to support preparation for the two tests and the final examination.

ASSESSMENT

Assessment structure

01

Quizzes

An Amathuba quiz will follow each pair of lectures. Each quiz will be released once the relevant lectures have been completed and will normally be due two or three days later.

02

Test 1

Test 1 will be a written test held during the first 45 minutes of the scheduled lecture on Friday 14 August.

03

Test 2

Test 2 will be a written test held after the second half of the course concludes on Monday 24 August. The date will be confirmed, but it is provisionally scheduled for Friday 28 August or Monday 31 August.

04

Final examination

The final examination will be a five-hour, computer-based examination. It will comprise a theory section, likely using multiple-choice questions, followed by a coding section completed in R.

FINAL MARK

Assessment weighting

60%

Final examination

15%

Test 1

15%

Test 2

10%

Quizzes